

Anurag Mahapatra

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Full-stack software engineer with hands-on experience in secure, distributed systems and AI-integrated applications. Maintained cloud infrastructure serving 3,000+ students, developed a real-time attendance platform to minimise proxy and engineered an LLM quality-assurance tool using a decomposed multi-model local pipeline. Delivers across the full stack from architecture and implementation through containerized deployment and CI/CD, with practical agile workflows, REST APIs, relational databases and AWS.

Technical Skills

Languages	Python, JavaScript, TypeScript, Java, C/C++, SQL, Bash
AI/ML	LangChain, RAG (FAISS), NLP (VADER, TF-IDF), Hugging Face, OpenAI & Anthropic SDKs
Web Technologies	React, Node.js, Express, FastAPI, MongoDB, PostgreSQL, WebSockets, REST APIs, JWT, Next.js
DevOps & Cloud	Docker, AWS (EC2, S3, RDS, Lambda, CloudWatch), Nginx, Redis, Kafka, GitHub Actions, CI/CD, Linux
Tools & Frameworks	Git, Cloudflare, AST Parsing (tree-sitter), OAuth 2.0, pytest, Jest, Agile/Scrum

Technical Projects

DQAS – Dynamic QR Attendance System GitHub

- Developed a full-stack attendance platform with a React faculty portal and React Native student app backed by Node.js and MongoDB; validated across 86 students in a live classroom setting with strong adoption feedback and active commercialization interest
- Engineered tamper-resistant QR generation using AES-256 encryption with 1-second refresh via WebSocket, plus device biometric gating and server-side token validation to eliminate proxy attendance; architecting a SaaS deployment path for institutional rollout

SYA Detector – LLM Sycophancy Detection Chrome Extension GitHub

- Created a Chrome extension that detects unjustified stance shifts in LLM responses with ~150–300 ms processing latency, intercepting outputs before final render across Claude, ChatGPT and Gemini
- Designed a two-model local inference pipeline where an Extractor handles stance extraction and new-information checks separately, then passes distilled summaries to a Judge model; reduces token usage by ~50–60% compared to single-model full-context approaches while maintaining consistent detection behavior

CommitSense – Intelligent Git Commit Quality Analyzer GitHub

- Developed an AST-based static analyzer for Python and JavaScript using tree-sitter to flag anti-patterns such as oversized diffs (>500 LOC), inconsistent naming and undocumented breaking changes, each scored by severity
- Integrated an LLM for semantic validation to align commit messages with code changes, cutting time spent writing meaningful commits by ~80%; containerized via Docker Compose with GitHub webhooks and per-repo quality dashboards

PromptCouncil – AI Debate Engine with Multi-Model Chat GitHub

- Engineered an AI orchestration engine that dynamically partitions available models into 2 opposing factions and drives structured multi-turn debates over a user-defined proposition, enforcing positional consistency across turns via a script-controlled moderator
- Shipped a multi-model chat interface with mid-conversation model switching across Anthropic/OpenAI-compatible APIs (including Ollama), preserving full context; unified abstraction layer normalises response formats and streaming behaviour across providers with ~200–400 ms switch latency

Experience

Data Analytics Intern – IBM SkillsBuild & CSRBOX June 2024 – August 2024

- Developed Python data processing pipelines (Pandas, NumPy) to clean and analyze real-world datasets of ~10k+ records; automated visualization workflows and delivered stakeholder-ready reports applying statistical analysis across multiple domain datasets

Education & Certifications

B.Tech in Computer Science & Engineering – CGPA: 8.8/10.0 Expected May 2027

Veer Surendra Sai University of Technology (VSSUT), Burla, Odisha

Relevant Coursework: Data Structures & Algorithms, Database Systems, Artificial Intelligence & Machine Learning, Operating Systems

Fundamentals of Accelerated Computing with CUDA Python July 2025

NVIDIA Deep Learning Institute

Leadership & Activities

Cloud & DevOps Lead, Enigma Web & Coding Club, VSSUT 2024 – Present

- Implemented and maintained CI/CD pipelines for veerpreps.com, a notes platform serving 3,000+ students during exam periods, using GitHub Actions and Docker; mentored 17 junior developers on containerization, AWS deployment and infrastructure best practices

Student Secretary, Department of Computer Science & Engineering, VSSUT 2025 – Present

- Lead planning and execution of departmental initiatives for 1,000+ students and faculty, covering academic coordination, technical events, industry engagement and student welfare across cross-functional teams